40) Four friends have been identified as suspects for an unauthorized access into a computer system. They have made statements to the investigating authorities. Alice said, "Carlos did it." John said, "I did not do it." Carlos said "Diana did it." Diana said, "Carlos lied when he said that I did it".

a) If the authorities also know that exactly one of the four suspects is telling the truth, who did it? Explain your reasoning.

b) If the authorities also know that exactly one is lying, who did it? Explain your reasoning.

Soln: p: Alice tells the truth q: John tells the truth r: Carlos tells the truth s: Diana tells the truth t: Carlos did the access u: John did the access v: Diana did the access

① $p \Leftrightarrow t$ is True ② $q \Leftrightarrow \neg u$ is True ③ $r \Leftrightarrow v$ is True ④ $s \Leftrightarrow (\neg r \wedge \neg v)$ is True ⑤ $r \Leftrightarrow \neg s$ is True

a) $(p \wedge \neg q \wedge \neg r \wedge \neg s) \vee (\neg p \wedge q \wedge \neg r \wedge \neg s) \vee (\neg p \wedge \neg q \wedge r \wedge \neg s) \vee (\neg p \wedge \neg q \wedge \neg r \wedge s)$ is True.

Let $(p \wedge \neg q \wedge \neg r \wedge \neg s)$ be True. By defn. of AND, p is True, $\neg q$ is True, $\neg r$ is True, $\neg s$ is True. By defn. of negation, q, r and s are respectively False. From ① and defn. of bi-conditional, t is True. From ⑤ and defn of biconditional, r is True. This is a contradiction. ∴ $\boxed{\therefore p \text{ is False}}$. From ① and defn. of bi-conditional, t is False. ∴ $\boxed{\text{Carlos did not do it.}}$ ¬Sure

Let $(\neg p \wedge q \wedge \neg r \wedge \neg s)$ be True. By defn. of AND, $\neg p$ is True, q is True, $\neg r$ is True, $\neg s$ is True. By defn. of negation, p, r and s are respectively False. From ② and defn. of bi-conditional, $\neg u$ is True. $\boxed{\therefore \text{John did not access}}$ From ③ and defn. of bi-conditional, v is False. $\boxed{\therefore \text{Diana did not access}}$. From ④ and defn. of bi-conditional, $(\neg r \wedge \neg v)$ is False. By defn. of negation, $\neg v$ is True. By defn. of AND, $\neg r$ is False. By defn. of negation, r is True.